B28: SQL SuperHero Program : Session 3

1 message

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PRimary Key and Unique Key :

KEY is KEY, which is used to uniquely identify the data. This is basic of all KEY.

Primary key --> PK
PK is unique, NOT NULL, required , never duplicate and used for uniquely identify the data in table.
every table have a max 1 PK.

```
create table empinfo(
id number primary key,
ename varchar2(20),
deptid number
);
```

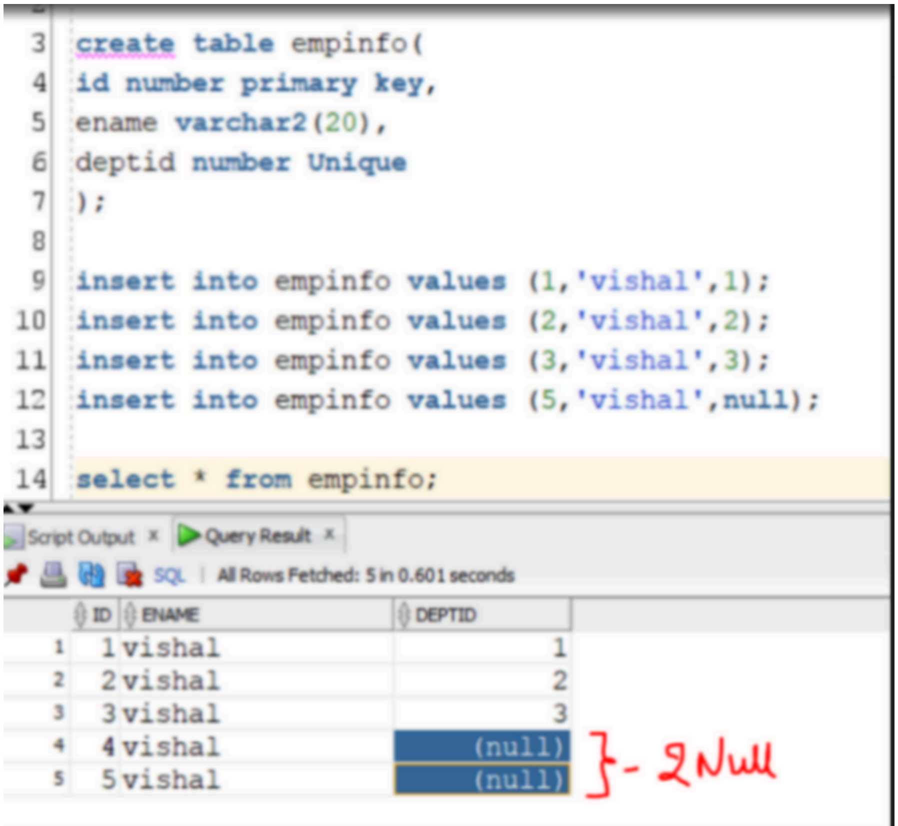
Unique key --> UK
UK is unique, required, never duplicate and used for uniquely identify the data in table.
table can have more than 1 unique key.
UK allows null.

every null is unique, we can not compare null to another null. thats why UK allows null, because it treats every null as unique.

10+null = null
10- null = null

null is unknown and undefined, we do not know the value, so it always gives us null.

```
create table empinfo(
id number primary key,
ename varchar2(20),
deptid number Unique
);
```



Null is allowed in the Unique key because every null is unique, as null is undefined and unknown. we can not compare one null to other null.

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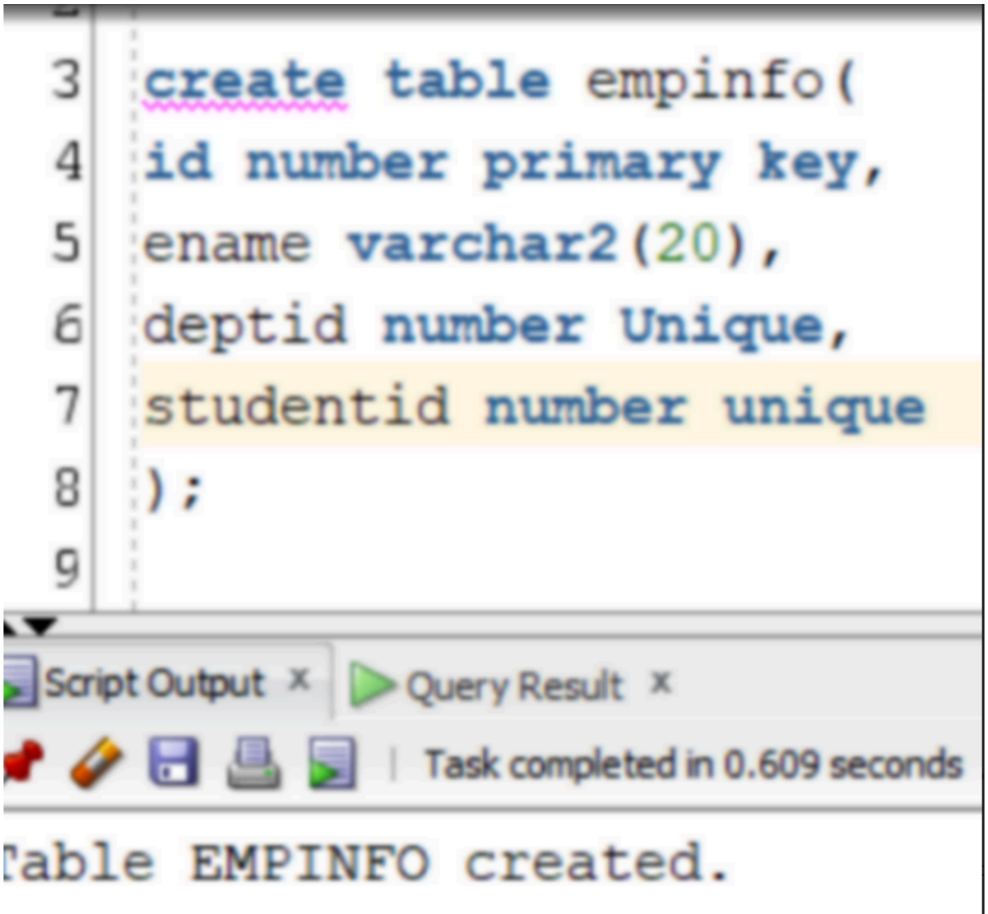
a table can have more than 1 unique key.

/* Remember the example we discussed on license number plate on a car is a primary key for RTO office/DMV office, but car have other unique values like engine number, VIN number etc.*/



composite key
A composite key is a key (primary or alternate) that consists of more than one column.

```
create table D (
 ID numeric(1),
 CODE varchar(2),
 constraint PK_D primary key (ID, CODE)
);
```

-> If there are 2 columns which can be used to uniquely identify the data in table, how can u decide which is PK and which is UK.?

We can see and understand the data which should come to the table.
1. If any one of those column can possibly come null, we can not take that column as PK because PK does not allow null, so we take that column as UK and another one as PK
2. if both of the columns will never null, then --> we can choose any of them as PK and prefer if anyone as NUMBER data type.

Question: In a composite Primary key, can any column which is a part of primary key can contain null value?

```
CREATE TABLE supplier
(
 supplier_id numeric(10) not null,
 supplier_name varchar2(50),
 contact_name varchar2(50),
 CONSTRAINT supplier_pk PRIMARY KEY (supplier_id, supplier_name)
);
```

--> In above example 2 column supplier_id, supplier_name is PK. Can I insert like below:

insert into supplier values (2,null,'c');

Answer: NO. None of the attribute is null

==

Question: Can I disable and enable the PK?

```
Answer: YES, we do, we can do disable the PK as well as enable.

ALTER TABLE supplier
enable CONSTRAINT supplier_pk;
```

===

Can a PK and a UK with not null constraint is same ?

YES

==

Can I create 2 PK on a table?

Obviously NO. But how to asnwer:
You could certainly mimick a second "primary" key by having an index placed on one or more other fields that are unique. Because what is PK ? PK is UK + Not Null constraint. PRIMARY KEY is usually equivalent to UNIQUE INDEX NOT NULL. So you can effectively have multiple "primary keys" on a single table.

```
create table test (id number primary key, name varchar2(20) unique not null);
```

Actually internally it behaves like 2 PK

Answer if you give NO there is no 2 PK that is the general answer, the way I am telling you if you give answer, actually you make an impact.

***From our class discussion today * Why Can not 2 PK**
Because Primary key is an identity to the row and there can't be two IDs against a row. If you remember B.Tech 2nd year the concept of Normalization of 2nd normal form and it says that Fully Functional dependency. It means If A, B --> C, so it is not possible that A --> C or B --> C. When we move to 3NF, by default everything is in 2NF for must, that is backward compatibility.
Now when answer the question , give your logic and add :
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create table test (id number primary key, name varchar2(20) unique not null);

Actually internally it behaves like 2 PK

Answer if you give NO there is no 2 PK that is general answer, the way I am telling you if you give answer, actually you make an impact. What to answer is not important, how to answer is more important.

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composite primary key Vs Sequence(Surrogate Key)

This is a controversial point. If my table with a composite primary key is expected to have millions of rows, the index controlling the composite key can grow up to a point where CRUD operation performance is very degraded. In that case, it is a lot better to use a simple integer ID primary key whose index will be compact enough and establish the necessary DBE constraints to maintain uniqueness.

===

Question: I have a table in production which has duplicate data. I can not delete existing rows because of business rules but somehow I have to prevent any further duplication of data in table. How to acheive it?

Yes, we can create a primary key on that column with a deferrable novalidate option.

```
create table xyz (id number);

insert into xyz values (1);
insert into xyz values (1);
insert into xyz values (1);
```

```
select * from xyz;
```

ID
1
1
1

≡ Duplicate data

Download CSV

3 rows selected.

If I want to add a PK constraint , it gives an error, because obviously PK not allow duplicate and this column already have duplicate.

```
alter table xyz add constraint pk_vishal  primary key (id)
```

→ Bcoz already duplicate data

```
ORA-02437: cannot validate (SQL_HBAMZQUXORBCJATTNIDGMXRDI.PK_VISHAL) - primary key violated
```

More Details: https://docs.oracle.com/error-help/db/ora-02437

Lets create with deferrable option:

```
alter table xyz add constraint pk_vishal  primary key (id) deferrable novalidate
```

```
Table altered.
```

✓ Done

```
insert into xyz values (1)
```

→ Now can not insert duplicate

```
ORA-00001: unique constraint (SQL_HBAMZQUXORBCJATTNIDGMXRDI.PK_VISHAL) violated ORA-06512: at "SYS.DBMS_SQL", line 1721
```

More Details: https://docs.oracle.com/error-help/db/ora-00001

deferrable novalidate option allow to create a PK without validating any old value. The strictions will be applicable for new data.